

Interface Capability Brief

HL7, EDI & FHIR Data Integration for Analytics & Reporting · Stakeholder overview · 2026-09-02

1. Executive summary

Show the eiConsole multi-source analytics route from the official video: HL7 over LLP, claims X12 over FTP, and FHIR JSON over REST, each normalized to a patient XML canonical, then written to a reporting table and posted as JSON.

2. Who this is for

- Sources: hospital ADT (LLP), provider 837 (FTP), cloud EMR FHIR Patient (REST) - Destinations: SQL Server AnalyticsDemo.dbo.PATIENT, mock quality-reporting REST - Demo vs production: synthetic samples only. Sibling HIMSS interfaces 01–03 are in eip-root so the console list matches the video open; only route 07 runs on EIP.

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

Integration highlights

FQCN sources: Jenny HIMSS 2018 route + PilotFish_V2 modules.conf. Image: pilotfish-eip:23R1. X12 tables: mount EDI/TableData/x12.

Stage	Module / mechanism	Notes
Listener	HL7TCPLListener / FTPLListener / RESTfulWebServiceListener	Ports 10001, FTP, /rest/FHIR
Source Transform	Formats HL7 to XML, EDI to XML, JSON-FHIR to XML	Wire module + Data Mapper ToXML
Routing	NullRoutingModule	Both targets
Target Transform	XML to DB (FromXML SQLXML), XML to JSON	Jenny maps
Transport	DatabaseSqlTransport, HttpPostTransport	\$\$\$db.*, \$\$\$simple.server.path

4. Honest boundaries (not claimed yet)

- None for this website slice.

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “Multiple Feeds to DB and API - HL7 - EDI - FHIR JSON” from the Web UI or documented smoke path

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- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- Checked: LLP ACK, PATIENT row, analytics JSON file - Not checked: full 837 IG, FHIR profile, SNIP

6. State & idempotency

- SQL insert is append-only. Resend duplicates (accepted).

8. Dual-write / side effects

- One inbound message → SQL insert and JSON POST

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

Multiple Feeds to DB and API - HL7 - EDI - FHIR JSON: runs steps such as HL7 Source System, FTP Listener, RESTful Listener, Snapshot SQL XML.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Risk noted in design: Med — EDI XML shape vs Jenny XSLT (2010AA billing provider) — Enhanced vs basic EDI XML
- Risk noted in design: Med — FHIR JSON → XML element names — JSON module 2.0 vs 2018 mapper
- Risk noted in design: Low — FTP PASV in Docker — EIP talks to ftp service

8. Seeing it work

- - Ports: Web UI 8154, EIP 8153, LLP 10001, FTP 2121, analytics 7072, SQL 14344 - Credentials: SQL sa / PilotFish_Demo1!; FTP demo / demo - Heap: 2GB - Cold start: SQL healthy + init, then EIP ~60–90s
- docker compose --profile full up -d --build
- - Local Web UI: http://127.0.0.1:8154/
- - EIP: http://127.0.0.1:8153/eip/
- - HL7 LLP: 127.0.0.1:10001
- - FTP: 127.0.0.1:2121 (demo / demo)
- - Analytics: http://127.0.0.1:7072/aggregationanalytics/restservice
- - SQL Server: 127.0.0.1:14344 (sa / PilotFish_Demo1!)

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.